

JAWAD UL HADI

Backend Lead Engineer | Backend Architecture | AI-First Systems Design

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PROFESSIONAL SUMMARY

Backend Lead Engineer with 8+ years of experience building distributed systems for multi-tenant SaaS and enterprise products. At MicroAgility, I lead backend architecture for Talentnix, a multi-tenant SaaS platform, and the APAC Management System (HRMS), and I own both from database schema to deployment. I work mostly in Node.js, NestJS, and TypeScript on PostgreSQL, MySQL, MongoDB, and Redis, with Python and FastAPI where they fit. My recent focus is production AI: Anthropic-based features, MCP servers, a provider-agnostic LLM layer, and RAG pipelines, built to keep failures contained. I mentor three associate developers and turn stakeholder requirements into agile delivery plans.

TECHNICAL SKILLS

Architecture & Systems Design: Distributed systems, system design, microservices, event-driven architecture, multi-tenant SaaS, REST and GraphQL API design, scalability and reliability trade-offs

Backend Engineering: Node.js, NestJS, TypeScript, Express.js, Python, FastAPI, Django REST Framework, MEAN stack, PostgreSQL, MySQL, MongoDB, Redis, BullMQ, MeiliSearch, query and performance optimisation

AI-First Engineering: LLM integration (OpenAI, Gemini, Anthropic), RAG pipelines, MCP servers, AI agent workflows, Claude Code, GitHub Copilot, Gemini 2.5 Flash

Cloud & Delivery: AWS (Lambda, API Gateway), GCP, OpenStack, Docker, Kubernetes, CI/CD (GitHub Actions), Jest, automated testing, OAuth 2.0/JWT

Product Domains: Multi-tenant SaaS, HRMS, CRM

Leadership: Technical leadership, mentoring and onboarding, architecture and code review, technical direction, architecture decision records (ADRs)

PROFESSIONAL EXPERIENCE

Backend Lead Engineer

MicroAgility Services (Pvt) Ltd. | Islamabad, Pakistan | Jan 2024 – Present

- Built the AI gateway for Talentnix: one routing layer in front of OpenAI, Gemini, and Anthropic with automatic failover. It cut AI integration complexity by 60%.
- Designed Talentnix's Retry → RAG Fallback → Rule-Based Floor pattern. Users have seen zero AI failures during live LLM outages (case study: "Designing for AI Failure").
- Enforced tenant isolation across Talentnix's REST and GraphQL APIs with OAuth 2.0/JWT. No cross-tenant data exposure incidents to date.
- Led backend development for the APAC Management System, MicroAgility's HRMS covering payroll, HR, attendance, and employee records, and for client projects.
- Cut dashboard response times from 12s to under 2s (83%) with PostgreSQL and MongoDB query tuning, new indexes, and a MeiliSearch rollout.
- Built an event-driven BullMQ and Redis job pipeline that keeps APIs under a second while thousands of async jobs run concurrently.
- Delivered full Jest unit-test coverage using GitHub Copilot, and built an internal ATS with Claude Code and Gemini 2.5 Flash.
- Set up GitHub Actions CI/CD pipelines with automated test gates, which shortened release cycles.
- Mentored three associate backend developers through the company's Learning & Development program.

Backend Software Engineer

MicroAgility Services (Pvt) Ltd. | Islamabad, Pakistan | Mar 2022 – Jan 2024

- Owned backend development for iAgility and AgileBrains, two large products built around the hiring and consultancy workflow, on a MEAN (MongoDB, Express.js, Angular, Node.js) microservices architecture.
- Built Express.js microservices and GraphQL APIs for each product's hiring flow, with MongoDB as the primary data store.
- Integrated Postal for transactional email, Google Cloud Storage for files, and WebSocket for real-time notifications.
- Ran zero-downtime schema migrations on live multi-tenant production databases while maintaining 99.9% uptime.

- Refactored the REST and GraphQL services used by thousands of users across multiple tenants, improving data retrieval speed by about 40%.
- Automated the deployment and migration pipeline for new enterprise tenants, cutting manual onboarding effort.

Software Engineer

Market Icon | Rawalpindi, Pakistan | Aug 2018 – Mar 2022

- Primary full-stack developer on a large-scale MEAN (MongoDB, Express.js, Angular, Node.js) system, with ownership of backend design and delivery.
- Designed the distributed server and database architecture on an OpenStack cloud platform.
- Modelled databases for continuously growing real-time data so reads and writes stayed fast as volume rose.
- Built serverless AWS Lambda and API Gateway services on an event-driven design, removing the idle cost of always-on servers.
- Built Express.js, FastAPI, and Django REST Framework APIs for secure CRM and payment integrations.
- Hardened backend services against peak load and power outages, and introduced code documentation and change tracking.

EDUCATION

B.S. Computer Science | Government College University, Faisalabad, Pakistan | Aug 2018

CERTIFICATIONS

- Claude Code in Action (Anthropic)
- Containers & Kubernetes Essentials (IBM)
- Gen AI: Beyond the Chatbot (Google)
- Introduction to Generative AI and Agents (Microsoft)
- Certified Django Developer

Full verified list: juh-bukhari.vercel.app

ADDITIONAL INFORMATION

Languages: English (professional working proficiency), Urdu (native)

Availability: Full-time. Remote from Islamabad with working hours aligned to EST/PST; also open to hybrid, on-site, or relocation with support.